

What is Discrete Mathematics?

Discrete mathematics is the part of mathematics devoted to the study of discrete (**as opposed to continuous**) objects.

Calculus deals with continuous objects and is not part of discrete mathematics.

Examples of discrete objects: integers, steps taken by a computer program, distinct paths to travel from point A to point B on a map along a road network, ways to pick a winning set of numbers in a lottery.

A course in discrete mathematics provides the mathematical background needed for **all subsequent courses in computer science** and for all subsequent courses in the many branches of discrete mathematics.

Kinds of Problems Solved Using Discrete Mathematics₁

How many ways can a password be chosen following specific rules?

How many valid Internet addresses are there?

What is the probability of winning a particular lottery?

Is there a link between two computers in a network?

How can I identify spam email messages?

How can I encrypt a message so that no unintended recipient can read it?

How can we build a circuit that adds two integers?

Kinds of Problems Solved Using Discrete Mathematics₂

What is the shortest path between two cities using a transportation system?

Find the shortest tour that visits each of a group of cities only once and then ends in the starting city.

How can we represent English sentences so that a computer can reason with them?

How can we prove that there are infinitely many prime numbers?

How can a list of integers be sorted so that the integers are in increasing order?

How many steps are required to do such a sorting?

How can it be proved that a sorting algorithm always correctly sorts a list?

Chapter 1.

The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs → 證明

1.8 Proof Methods and Strategy → 證明的方法&策略

1-1
的
應用

Propositions (命題)

- A **proposition (命題)** is a declarative sentence (陳述句) that is either true or false. → 命題是一個要嘛對或錯的陳述句

➤ Examples of propositions: ↓是個「命題」

1、定義 Definition ⇒ 背!

2、用正例、反例強化

- a) The Moon is made of green cheese. (false)
- b) Trenton is the capital of New Jersey. (true)
- c) Toronto is the capital of Canada. (false)
- d) $1 + 0 = 1$ (true)
- e) $0 + 0 = 2$ (false)

➤ Examples that are not propositions. ↓不是「命題」(沒有對錯的解)

- f) Sit down!
- g) What time is it?
- h) $x + 1 = 2$
- i) $x + y = z$

Propositional Logic (命題邏輯)

Constructing Propositions

- Propositional Variables: $p, q, r, s, \dots$
- The proposition that is always true is denoted by **T** and the proposition that is always false is denoted by **F**.
- Compound Propositions (複合命題); constructed from logical connectives and other propositions



- **Negation** $\neg$
- **Conjunction** $\wedge$
- **Disjunction** $\vee$
- **Implication** $\rightarrow$
- **Biconditional** $\leftrightarrow$

Negation 否定

< Definition >

The **negation** of a proposition p is denoted by $\neg p$ and has this truth table:

p	$\neg p$
T	F
F	T

Example: If p denotes “The earth is round.”, then $\neg p$ denotes “It is not the case that the earth is round,” or more simply “The earth is **not** round.”

Conjunction



Definition 2

Let p and q be propositions. The *conjunction* of p and q , denoted by $p \wedge q$, is the proposition “ p and q .” The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

The *conjunction* of propositions p and q is denoted by $p \wedge q$ and has this truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \wedge q$ denotes “I am at home **and** it is raining.”

Disjunction

or 或

The *disjunction* of propositions p and q is denoted by $p \vee q$ and has this truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \vee q$ denotes “I am at home **or** it is raining.”

The Connective Or in English

In English “or” has two distinct meanings.



• **“Inclusive Or”** - In the sentence “Students who have taken CS202 or Math120 may take this class,” we assume that students need to have taken one of the prerequisites, but may have taken both. This is the meaning of disjunction. For $p \vee q$ to be true, **either one or both** of p and q must be true.



• **“Exclusive Or”** - When reading the sentence “Soup or salad comes with this entrée,” we **do not expect to be able to get both** soup and salad. This is the meaning of Exclusive Or (**xor**). In $p \oplus q$, one of p and q must be true, but not both. The truth table for $\oplus$ is:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

只有一個成立
才為真！



Implication

暗示、涵意
若……則……

Definition 5 Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition “if p , then q .” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the *hypothesis* (or *antecedent* or *premise*) and q is called the *conclusion* (or *consequence*).

If p and q are propositions, then $p \rightarrow q$ is a **conditional statement** (條件語句) or **implication** which is read as “if p , then q ” and has this truth table:

$p \rightarrow q$ 等價於 $(\neg p) \vee q$

p	$\neg p$	q	$(\neg p) \vee q$	$p \rightarrow q$
T	F	T	T	T
T	F	F	F	F
F	T	T	T	T
F	T	F	T	T

可以發現 $p \rightarrow q$ 相當於 $(\neg p) \vee (q)$
NOT p or q
前面對後面就一定要對

唯一 False

注意， $p \rightarrow q$ 僅表示：如果 p 是 true，則 q 是 true。沒有說如果 p 是 false 的情形。前面錯後面不一定錯！

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \rightarrow q$ denotes “**If** I am at home **then** it is raining.”

In $p \rightarrow q$, p is the *hypothesis* (antecedent or premise) and q is the *conclusion* (or consequence).

Understanding Implication₁

In $p \rightarrow q$ there does not need to be any connection between the antecedent or the consequent. **The “meaning” of $p \rightarrow q$ depends only on the truth values of p and q .**

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

These implications are perfectly fine, but would not be used in ordinary English.

- “If the moon is made of green cheese, then I have more money than Bill Gates.” $\rightarrow$ True

- “If the moon is made of green cheese then I’m on welfare.” $\rightarrow$ True

- “If $1 + 1 = 3$, then your grandma wears combat boots.” $\rightarrow$ True

前提錯的情況下，後句永遠用不到（無法利用 p 去 proof q ），但亦無法反駁 q ，所以最終結論為 True。

Understanding Implication₂

One way to view the logical conditional is to think of an obligation or contract.

- “If **I am elected**, then **I will lower taxes**.”
- “If you get 100% on the final, then you will get an A.”

If the politician **is elected** and **does not lower taxes**, then the voters can say that he or she has broken the campaign pledge. Something similar holds for the professor. This corresponds to the case where p is true and q is false.

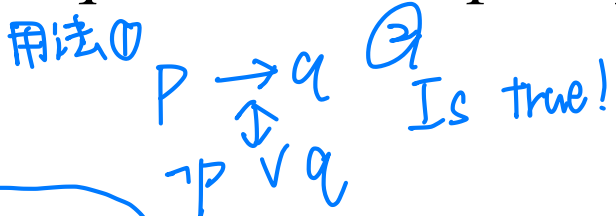
Different Ways of Expressing $p \rightarrow q$

注意：以下都是 $p \rightarrow q$ 的同一種意思！

if p , then q

p implies q

if p , q



q unless $\neg p$

q when p

q if p

p only if q

q is true unless $\neg p$ is true

特別注意這兩個

記住： $p \rightarrow q$ 表示：

1. 若 p 是 true，則 q 是 true

2. 若 q 是 false，則 p 是 false

舉例： p : 今天下雨
 q : 帶傘出門

q whenever p

q follows from p

高中

p is sufficient for q (p 是 q 的充分條件)

q is necessary for p (q 是 p 的必要條件)

a necessary condition for p is q

a sufficient condition for q is p

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Converse, Contrapositive, and Inverse

相反的 逆否 对立

From $p \rightarrow q$ we can form new conditional statements .

- $q \rightarrow p$ is the **converse** of $p \rightarrow q$
- $\neg q \rightarrow \neg p$ is the **contrapositive (逆否)** of $p \rightarrow q$
- $\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$

Example: Find the converse, inverse, and contrapositive of “It raining is a sufficient condition for my not going to town.” (若下雨則不進城)

Solution:

converse: If I do not go to town, then it is raining.

contrapositive: If I go to town, then it is not raining.

inverse: If it is not raining, then I will go to town.

p if and only if q **Biconditional** 若且為若
 $p \leftrightarrow q$ (iff)
 $(p \rightarrow q) \wedge (q \rightarrow p)$

If p and q are propositions, then we can form the **biconditional** proposition $p \leftrightarrow q$, read as “ **p if and only if q .**” The biconditional $p \leftrightarrow q$ denotes the proposition with this truth table:

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	F
F	F	T	T	T

If p denotes “I am at home.” and q denotes “It is raining.” then $p \leftrightarrow q$ denotes “I am at home **if and only if** it is raining.”

if and only if 簡寫為
iff

Expressing the Biconditional

Some alternative ways “ p if and only if q ” is expressed in English:

- p is necessary and sufficient for q p是q的充分必要條件
- if p then q , and conversely
- p iff q
- p exactly when (與某事同時發生) q

Truth Tables For Compound Propositions

複合

Construction of a truth table:

Rows 列 →

- Need a row for **every possible combination** of values for the atomic propositions.

Columns 欄、行 ↓

- Need a column **for the compound proposition** (usually at far right)
- Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
 - This includes the **atomic propositions** (無法拆成更小命題的)

Example Truth Table

$p \rightarrow q$ 相當於 $\neg p \vee q$

Construct a truth table for $(p \vee q) \rightarrow (\neg r)$

Columns 行、欄

! 注意優先順序

$p \vee q \rightarrow \neg r$ 相當於 $\neg (p \vee q) \vee (\neg r)$

Row	p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
列						
	T	T	T	F	T	F
	T	T	F	T	T	T
	T	F	T	F	T	F
	T	F	F	T	T	T
	F	T	T	F	T	F
	F	T	F	T	T	T
	F	F	T	F	F	T
	F	F	F	T	F	T

Example Truth Table

$p \rightarrow q$ 相當於 $\neg p \vee q$

Construct a truth table for $p \vee q \rightarrow \neg r$

$p \vee q \rightarrow \neg r$ 相當於 $\neg (p \vee q) \vee (\neg r)$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Equivalent Propositions

Two propositions are equivalent (等價) if they **always have the same truth value**.

Example: Show using a truth table that the conditional is **equivalent to the contrapositive**.

Solution:

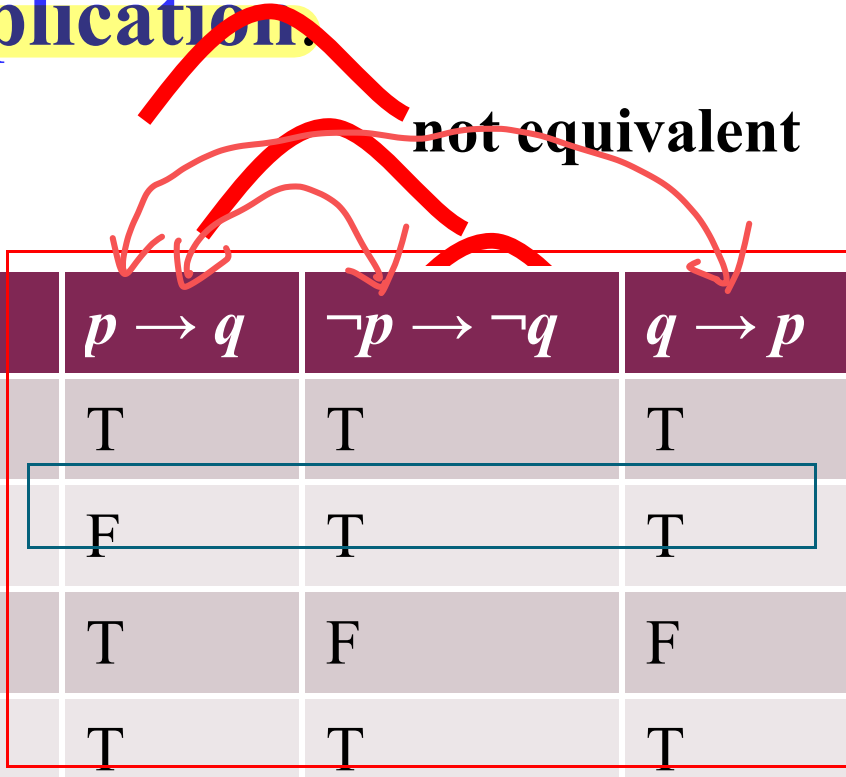
等價 (等) equivalent

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Using a Truth Table to Show Non-Equivalence

Example: Show using truth tables that neither the converse nor inverse of an implication are not equivalent to the implication.

Solution:



p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

Problem

How many rows are there in a truth table with n propositional variables?

like p, q : Row 4 (2^2)
 p, q, r : Row 8 (2^3)

Solution: 2^n

(We will see how to do this in Chapter 6.)

Note that this means that with n propositional variables, **we can construct 2^n distinct** (that is, not equivalent) **propositions**.

Precedence of Logical Operators

(符號的優先順序)

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Java Boolean Logical Operator		
Priority	Operator	Simple Name
1	!	Logical Unary NOT
2	&	Logical AND
3	^	Exclusive OR or XOR
4		Logical OR
5	&&	Logical Short Circuit AND
6		Logical Short Circuit OR
7	op=	Compound Assignment Operator &=, =, ^=
7	=	Assignment

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow \neg r$

Note: If the intended meaning is $p \vee (q \rightarrow \neg r)$, then parentheses must be used.

XOR has different position in languages (e.g., C, C++, Java), but it has surely not higher precedence than AND and not lower than OR.

↑ 確定不高於且、不低於或

Chapter 1.

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1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

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1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Translating English Sentences

Steps to convert an English sentence to a statement in propositional logic

- Identify atomic propositions and represent using propositional variables.
- Determine appropriate logical connectives

“**If I go to Harry’s or to the country, I will not go shopping.**”

- p : I go to Harry’s
- q : I go to the country.
- r : I will go shopping.

*根據自己的習慣來定義 😊

If p or q then not r .

$$(p \vee q) \rightarrow \neg r$$

Example

Problem: Translate the following sentence into propositional logic:

“You can access the Internet from campus **only if** you are a computer science major **or** you are **not** a freshman.”

One solution: Let a , c , and f represent

- a : You can access the internet from campus
 - c : You are a computer science major
 - f : You are a freshman.”
- Handwritten notes:* "only if" with an arrow pointing to the first bullet, and "or" with a bracket pointing to the second and third bullets.

$$a \rightarrow (c \vee \neg f)$$

System Specifications

System and Software engineers take requirements in English and express them in a precise specification language based on logic.

Example: Express in propositional logic:

“The automated reply **cannot** be sent **when** the file system is full”

One solution:

- p : The automated reply can be sent
- q : The file system is full

$$q \rightarrow \neg p$$

note P.6
 q when p
only if

Consistent System Specifications

Definition: A list of propositions is **consistent** (一致) if it is possible to assign truth values to the proposition variables so that each proposition is true. (存在一種解使得全部都是true) (有解)

要找到!

Exercise: Are these specifications consistent?

- “The diagnostic message is stored in the buffer or it is retransmitted.” $\neg p \vee q$
 - “The diagnostic message is not stored in the buffer.” $\neg p$
 - “If the diagnostic message is stored in the buffer, then it is retransmitted.” $p \rightarrow q$
- 診斷 重發重傳

Solution: Let p denote “The diagnostic message is stored in the buffer.” Let q denote “The diagnostic message is retransmitted.” The specification can be written as: $p \vee q$, $\neg p$, $p \rightarrow q$. **When p is false and q is true all three statements are true.** So the specification is consistent.

- What if “The diagnostic message is not retransmitted” is added.

Solution: Now we are adding $\neg q$ and there is no satisfying assignment. So the specification is not consistent.

Logic Puzzles



Raymond Smullyan
(Born 1919)

An island has two kinds of inhabitants, knights (騎士), who always tell the truth, and knaves (惡棍), who always lie.

You go to the island and meet A and B.

解法：

- A says “B is a knight.”

- p : A is a knight

- q : B is a knight

- B says “The two of us are of opposite types. (兩者不同類型的)”

Example: What are the types of A and B?

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively. So, then $\neg p$ represents the proposition that A is a knave and $\neg q$ that B is a knave.

分別

- If A is a knight, then p is true. Since knights tell the truth, q must also be true. Then $(p \wedge \neg q) \vee (\neg p \wedge q)$ would have to be true, but it is not. So, A is not a knight and therefore $\neg p$ must be true.

- If A is a knave, then B must not be a knight since knaves always lie. So then both $\neg p$ and $\neg q$ hold since both are knaves. **Ans: both A and B are knaves.**

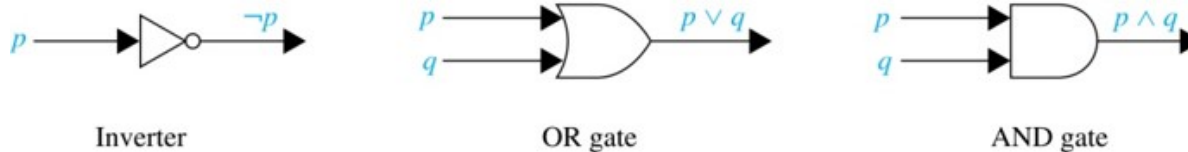
Logic Circuits

(Studied in depth in Chapter 12)

Electronic circuits; each input/output signal can be viewed as a 0 or 1.

- 0 represents **False**
- 1 represents **True**

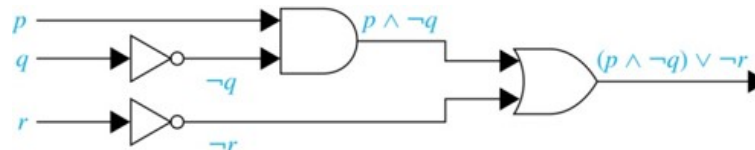
Complicated circuits are constructed from **three basic circuits** called gates.



- The inverter (**NOT gate**) takes an input bit and produces the negation of that bit.
- The **OR gate** takes two input bits and produces the value equivalent to the disjunction of the two bits.
- The **AND gate** takes two input bits and produces the value equivalent to the conjunction of the two bits.

More complicated digital circuits can be constructed by combining these basic circuits to produce the desired output given the input signals by building a circuit for each piece of the output expression and then combining them.

For example:



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Tautologies, Contradictions, and Contingencies

A **tautology** (恒真) is a proposition which is **always true**.

- Example: $p \vee \neg p$

A **contradiction** (矛盾) is a proposition which is **always false**.

- Example: $p \wedge \neg p$

A **contingency** (適真) is a proposition which is **neither a tautology nor a contradiction**, such as p

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logically Equivalent

Two compound propositions p and q are **logically equivalent (邏輯等價)** if $p \leftrightarrow q$ is a tautology.

We write this as ~~$p \leftrightarrow q$ or as $p \equiv q$~~ where p and q are compound propositions.
~~“ $p \leftrightarrow q$ 恒真”可直接寫成 $p \leftrightarrow q$ 或 $p \equiv q$ ”~~

Two compound propositions **p and q are equivalent** if and only if the columns in a **truth table giving their truth values agree.**

This truth table shows that ~~$\neg p \vee q$ is equivalent to $p \rightarrow q$.~~

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$	$(p \rightarrow q) \leftrightarrow (\neg p \vee q)$
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

De Morgan's Laws 迪摩根定理

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$



Augustus De Morgan
1806-1871

This truth table shows that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Key Logical Equivalences₁

依靠 Truth Table 來 proof!

p	$p \vee$
	F
T	T
F	F

Identity Laws:

(恆等律)

$$p \wedge T \equiv p,$$

$$p \vee F \equiv p$$

Domination

Laws: (支配律)

$$p \vee T \equiv T,$$

$$p \wedge F \equiv F$$

Idempotent laws:

(冪等律)

$$p \vee p \equiv p,$$

$$p \wedge p \equiv p$$

Double Negation

Law: (雙非律)

$$\neg(\neg p) \equiv p$$

Negation Laws:

(否定律)

$$p \vee \neg p \equiv T,$$

$$p \wedge \neg p \equiv F$$

Key Logical Equivalences₂

Commutative
Laws: (交換律)

$$p \vee q \equiv q \vee p, \quad p \wedge q \equiv q \wedge p$$

Associative Laws:
(結合律)

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

Distributive
Laws: (分配律)

$$(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$$

$$(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$$

Absorption Laws:
(吸收律)

$$p \vee (p \wedge q) \equiv p \quad p \wedge (p \vee q) \equiv p$$

$$\neg (p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg (p \vee q) \equiv \neg p \wedge \neg q$$

 De Morgan's
Laws

* 用 Truth Table 作一遍, 會!

More Logical Equivalences

TABLE 7 Logical Equivalences
Involving Conditional Statements.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

TABLE 8 Logical Equivalences
Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Constructing New Logical Equivalences

We can show that two expressions are logically equivalent by developing a series of logically equivalent statements.

To prove that $A \equiv B$ we produce a series of equivalences beginning with A and ending with B.

$$\begin{array}{c} A \equiv A_1 \\ \boxed{?} \\ A_n \equiv B \end{array}$$

Keep in mind that whenever a proposition (represented by a propositional variable) occurs in the equivalences listed earlier, it may be replaced by an arbitrarily complex compound proposition.

Equivalence Proofs₁

Example: Show $\neg(p \vee (\neg p \wedge q))$
that is logically equivalent to $\neg p \wedge \neg q$

用定理 / 現有條件
↓
proof

Solution:

$$\begin{aligned}\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the second De Morgan law} \\ &\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the first De Morgan law} \\ &\equiv \neg p \wedge (p \vee \neg q) && \text{law by the double negation law} \\ &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the second distributive law} \\ &\equiv F \vee (\neg p \wedge \neg q) && \text{because } \neg p \wedge p \equiv F \\ &\equiv (\neg p \wedge \neg q) \vee F && \text{by the commutative law for disjunction} \\ &\equiv (\neg p \wedge \neg q) && \text{By the identity law for F}\end{aligned}$$

Equivalence Proofs₂

Example: Show $(p \wedge q) \rightarrow (p \vee q)$
that is a tautology (恒真)

Solution:

$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{by truth table for } \rightarrow \\ &\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{by the first De Morgan law} \\ &\equiv (\neg p \vee p) \vee (\neg q \vee q) && \text{by associative and commutative laws} \\ &&& \text{laws for disjunction} \\ &\equiv T \vee T && \text{by truth tables} \\ &\equiv T && \text{by the domination law}\end{aligned}$$

Propositional Satisfiability

複合

A compound proposition is *satisfiable* (可滿足的) if there is an assignment of truth values to its variables that make it true. When no such assignments exist, the compound proposition is *unsatisfiable*.

不可滿足

A compound proposition is *unsatisfiable* if and only if its negation is a tautology.

Questions on Propositional Satisfiability

(make it true)

Example: Determine the satisfiability of the following compound propositions:

Question $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$

Solution: Satisfiable. Assign **T** to p , q , and r .

Question $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$

Solution: Satisfiable. Assign **T** to p and **F** to q .

Question

$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$

Solution: Not satisfiable. (Check each possible assignment of truth values to the propositional variables and none will make the proposition true.)

Notation

$\bigvee_{j=1}^n p_j$ is used for $p_1 \vee p_2 \vee \dots \vee p_n$

$\bigwedge_{j=1}^n p_j$ is used for $p_1 \wedge p_2 \wedge \dots \wedge p_n$

Needed for the next example.

Sudoku

A **Sudoku puzzle** is represented by a 9×9 grid made up of nine 3×3 subgrids, known as **blocks**. Some of the 81 cells of the puzzle are assigned one of the numbers 1,2, ..., 9.

The puzzle is solved by assigning numbers to each blank cell so that every row, column and block contains each of the nine possible numbers.

Example

	2	9				4		
			5		4	1		
	4	5						
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

Encoding as a Satisfiability Problem₂

For each cell with a given value, assert $p(i,j,n)$, when the cell in row i and column j has the given value.

Assert that **every row contains every number.**

$$\bigwedge_{i=1}^9 \bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$$

Assert that **every column contains every number.**

$$\bigwedge_{j=1}^9 \bigwedge_{n=1}^9 \bigvee_{i=1}^9 p(i, j, n)$$

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

Encoding as a Satisfiability Problem₃

Assert that each of the 3×3 blocks contain every number.

$$\bigwedge_{r=0}^2 \bigwedge_{s=0}^2 \bigwedge_{n=1}^9 \bigvee_{\bar{i}=1}^3 \bigvee_{\bar{j}=1}^3 p(3r+\bar{i}, 3s+\bar{j}, n)$$

(this is tricky - ideas from chapter 4 help)

Assert that no cell contains more than one number.
Take the conjunction over all values of n, n', i , and j , where each variable ranges from 1 to 9 and $n \neq n'$, of

$$p(i, j, n) \rightarrow \neg p(i, j, n')$$

Solving Satisfiability Problems

To solve a Sudoku puzzle, we need to find an assignment of truth values to the 729 variables of the form $p(i,j,n)$ that makes the conjunction of the assertions true. Those variables that are assigned T yield a solution to the puzzle.

A **truth table** can always be used to determine the satisfiability of a compound proposition. But this **is too complex even for modern computers for large problems**.

There has been much work on developing efficient methods for solving satisfiability problems as many practical problems can be translated into **satisfiability problems**.

Chapter 1.

The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Propositional Logic Not Enough

所有、存在 $\Rightarrow$ 沒有辦法利用命題邏輯
(至少有一個)

If we have:

凡人
“All men are mortal.”

所有
“Socrates is a man.”

阿蘇

阿蘇 是 凡人

Does it follow that “Socrates is mortal?”

Can't be represented in propositional logic.

Need a language that talks about objects, their properties, and their relations.

Later we'll see how to draw inferences.

Introducing Predicate Logic

Predicate logic (謂詞邏輯) uses the following new features:

- Variables: x, y, z
- Predicates: $P(x), M(x)$
- Quantifiers (*to be covered in a few slides*):

Propositional functions (命題函數) are a generalization of propositions.

- They contain variables and a predicate, e.g., $P(x)$
- Variables can be replaced by elements from their *domain*.

Propositional Functions 命題函數

Propositional functions become propositions (and have truth values) when their variables are each replaced by a value from the **domain** (or **bound** by a quantifier, as we will see later).
定義域

The statement $P(x)$ is said to be the value of the propositional function P at x .

For example, let $P(x)$ denote “ $x > 0$ ” and the domain be the integers. Then:

$P(-3)$ is false. $\rightarrow -3 > 0$ $P(0.1) \Rightarrow$ 未定義

$P(0)$ is false. $\rightarrow 0 > 0$ (Domain 是 INT)

$P(3)$ is true. $\rightarrow 3 > 0$

Often the domain is denoted by U . So in this example U is the integers.

Examples of Propositional Functions

Let “ $x + y = z$ ” be denoted by $R(x, y, z)$ and U (for all three variables) be the integers. Find these truth values:

$$R(2, -1, 5)$$

Solution: F

$$2 + (-1) = 5 \quad \text{false}$$

$$R(3, 4, 7)$$

Solution: T

$$3 + 4 = 7 \quad \text{True}$$

$$R(x, 3, z)$$

Solution: Not a Proposition

$$x + 3 = z \rightarrow \text{不是个命题 (没有 T or F)}$$

Now let “ $x - y = z$ ” be denoted by $Q(x, y, z)$, with U as the integers. Find these truth values:

$$Q(2, -1, 3)$$

Solution: T

$$2 - (-1) = 3 \quad \text{True}$$

$$Q(3, 4, 7)$$

Solution: F

$$3 - 4 = 7 \quad \text{False}$$

$$Q(x, 3, z)$$

Solution: Not a Proposition

$$x - 3 = z \rightarrow \text{不是个命题 (没有 T or F)}$$

Compound Expressions

Connectives from propositional logic carry over to predicate logic.

If $P(x)$ denotes " $x > 0$," find these truth values:

$$P(3) \vee P(-1)$$

$T \vee F$

Solution: T

$$P(3) \wedge P(-1)$$

$T \wedge F$

Solution: F

$$P(3) \rightarrow P(-1)$$

$T \rightarrow F$

Solution: F

$$P(3) \rightarrow \neg P(-1)$$

$T \rightarrow T$

Solution: T

→ " $3 > 0$ " $\vee$ " $-1 > 0$ "

Expressions with variables are not propositions and therefore do not have truth values. For example,

$$P(3) \wedge P(y)$$

$$P(x) \rightarrow P(y)$$

) 未定義

✱ When used with quantifiers (限定詞) (to be introduced next), these expressions (propositional functions) become propositions.

Quantifiers



Charles Peirce
(1839-1914)

We need **quantifiers (限定詞)** to express the meaning of English words **including all and some**:

- “**All** men are Mortal.”
- “**Some** cats do not have fur.”

(1) 都等價
 $\forall x, P(x)$
 $\forall x P(x)$
 $\forall x, P(x)$ is true

(2) 都等價
 $\exists x, P(x)$
 $\exists(x) P(x)$
 $\exists(x), P(x)$ is true

The two most important quantifiers are:

- **Universal Quantifier**, “For all,” symbol: $\forall$ (對於所有的)
- **Existential Quantifier**, “There exists,” symbol: $\exists$ (存在)

We write as in $\forall x P(x)$ and $\exists x P(x)$.

$\forall x P(x)$ asserts $P(x)$ is true for every x in the domain. 對於所有

$\exists x P(x)$ asserts $P(x)$ is true for some x in the domain. 對於部分

The quantifiers are said to bind the variable x in these expressions.

Universal Quantifier

$\forall x P(x)$ is read as "For all x , $P(x)$ " or "For every x , $P(x)$."

Examples:

- 1) If $P(x)$ denotes " $x > 0$ " and U is the integers, then $\forall x P(x)$ is false.
所有整數
- 2) If $P(x)$ denotes " $x > 0$ " and U is the positive integers, then $\forall x P(x)$ is true.
所有正整數
- 3) If $P(x)$ denotes " x is even" and U is the integers, then $\forall x P(x)$ is false.
 x 是偶數 對於所有整數

→ 處理 false 較簡單

Existential Quantifier

— 有沒有任何一個

$\exists x P(x)$ is read as “For some x , $P(x)$ ”, or as “There is an x such that $P(x)$,” or “For at least one x , $P(x)$.”

Examples:

1. If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\exists x P(x)$ is true. It is also true if U is the positive integers.
2. If $P(x)$ denotes “ $x < 0$ ” and U is the positive integers, then $\exists x P(x)$ is false.
3. If $P(x)$ denotes “ x is even” and U is the integers, then $\exists x P(x)$ is true.

→ 處理 true 較簡單

Thinking about Quantifiers

When the domain of discourse is **finite (有限的)**, we can think of quantification as **looping through the elements of the domain**.

To evaluate $\forall x P(x)$ loop through all x in the domain.

- If at **every** step $P(x)$ is true, then $\forall x P(x)$ is true.
- If at **a step $P(x)$ is false, then $\neg \forall x P(x)$ is false** and the loop terminates.

To evaluate $\exists x P(x)$ loop through all x in the domain.

- If at **some** step, $P(x)$ is true, then $\exists x P(x)$ is true and the loop terminates.
- If the loop ends **without finding an x for which $P(x)$ is true, then $\neg \exists x P(x)$ is false**.

Even **if the domains are infinite**, we can still think of the quantifiers this fashion, but the loops will not terminate in some cases.

Properties of Quantifiers

The truth value of $\exists x P(x)$ and $\forall x P(x)$ depend on both the propositional function $P(x)$ and on the domain U .

Examples:

存在(找到)! $\leftarrow$

! Domain 很重要!

1. If U is the positive integers and $P(x)$ is the statement " $x < 2$ ", then $\exists x P(x)$ is true, but $\forall x P(x)$ is false.
 $1 < 2 \rightarrow T$ $1 < 2$, but $3 > 2 \rightarrow F$
2. If U is the negative integers and $P(x)$ is the statement " $x < 2$ ", then both $\exists x P(x)$ and $\forall x P(x)$ are true.
組成 $-1 \rightarrow T$ $\forall$ all negative int < 2 (T)
3. If U consists of 3, 4, and 5, and $P(x)$ is the statement " $x > 2$ ", then both $\exists x P(x)$ and $\forall x P(x)$ are true. But if $P(x)$ is the statement " $x < 2$ ", then both $\exists x P(x)$ and $\forall x P(x)$ are false.

* consists 組成 * contain 包含 $\rightarrow$ 注意這兩符號

Precedence of Quantifiers

最高的優先順序!

The quantifiers $\forall$ and $\exists$ have higher precedence than all the logical operators.

For example, $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$

$\forall x (P(x) \vee Q(x))$ means something different.

Unfortunately, often people write $\forall x P(x) \vee Q(x)$ when they mean $\forall x (P(x) \vee Q(x))$.

Translating from English to Logic₁

找 Domain!

Example 1: Translate the following sentence into predicate logic: “Every student in this class has taken a course in Java.”

Solution:

First decide on the domain U .

∴ 這表示 : all people are students in this class and have taken a course in Java.

Solution 1: If U is all students in this class, define a propositional function $J(x)$ denoting “ x has taken a course in Java” and translate as $\forall x J(x)$.

Solution 2: But if U is all people, also define a propositional function $S(x)$ denoting “ x is a student in this class” and translate as $\forall x (S(x) \rightarrow J(x))$.

$\forall x (S(x) \wedge J(x))$ is not correct. What does it mean?

Translating from English to Logic₂

(1) 存在一个 x , 然后 $S(x) \wedge J(x)$

(2) 若是 $\exists x (S(x) \rightarrow J(x))$ 那这写法同时还等质 $(\exists x \neg S(x) \vee J(x))$ 思考! 还和原句一样吗?

Example 2: Translate the following sentence into predicate logic: "Some student in this class has taken a course in Java."

Solution:

First decide on the domain U .

Solution 1: If U is all students in this class, translate as

$$\exists x J(x)$$

$S(x)$ denoting "x is a student in this class"

$J(x)$ denoting "x has taken a course in Java"

Solution 2: But if U is all people, then translate as

(1) $\exists x (S(x) \wedge J(x))$ → 已成的既定事实, 所以为 $\wedge$ (已经有结果了)

(2) $\exists x (S(x) \rightarrow J(x))$ is not correct. What does it mean?

假设句!

∴ 只要前提不成立这个即为 true
例如: some people is not a student in this class.

Returning to the Socrates (蘇格拉底)

Example

Introduce the propositional functions *Man(x)* denoting “*x* is a man” and *Mortal(x)* denoting “*x* is mortal.”

Specify the domain as all people.

The two premises are: $\forall x (Man(x) \rightarrow Mortal(x))$

$Man(Socrates)$

The conclusion is: $Mortal(Socrates)$

Later we will show how to prove that the conclusion follows from the premises.

假设

Equivalences in Predicate Logic

Statements involving predicates and quantifiers are logically equivalent if and only if they have the same truth value

- for every predicate substituted into these statements and
- for every domain of discourse used for the variables in the expressions.

The notation $S \equiv T$ indicates that S and T are logically equivalent.

Example: $\forall x \neg \neg S(x) \equiv \forall x S(x)$

Thinking about Quantifiers as Conjunctions and Disjunctions

If the **domain is finite**, a **universally** quantified proposition is equivalent to a **conjunction** of propositions without quantifiers and an **existentially** quantified proposition is equivalent to a **disjunction** of propositions without quantifiers.

If U consists of the integers 1, 2, and 3:

$$\neg(\forall x P(x)) \equiv \neg(P(1) \wedge P(2) \wedge P(3)) \equiv (\neg P(1)) \vee (\neg P(2)) \vee (\neg P(3))$$

for All: $\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$ $\neg(\neg P(x))$

Exist: $\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$ $\exists x (\neg P(x))$

Even if the domains are infinite, you can still think of the quantifiers in this fashion, but the equivalent expressions without quantifiers will be infinitely long.

6 Negating Quantified Expressions₁

Consider $\forall x J(x)$

"**Every** student in your class has taken a course in Java."

Here $J(x)$ is "x has taken a course in Java" and
the domain is students in your class.

Negating the original statement gives "It is **not** the case that every student in your class has taken Java." This implies that "**There is a** student in your class who has **not** taken Java."

Symbolically $\neg \forall x J(x)$ and $\exists x \neg J(x)$ are equivalent

$$\neg \forall x J(x) \equiv \exists x \neg J(x)$$

Negating Quantified Expressions₂

Now Consider $\exists x J(x)$

"There is a student in this class who has taken a course in Java."

Where $J(x)$ is "x has taken a course in Java."

Negating the original statement gives "It is **not** the case that there is a student in this class who has taken Java." This implies that "Every student in this class has **not** taken Java" ✗

Symbolically $\neg \exists x J(x)$ and $\forall x \neg J(x)$ are equivalent

$$\neg \exists x J(x) \equiv \forall x \neg J(x)$$

De Morgan's Laws for Quantifiers

The rules for negating quantifiers are:

TABLE 2 De Morgan's Laws for Quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
<u>$\neg \exists x P(x)$</u>	<u>$\forall x \neg P(x)$</u>	For every x , $P(x)$ is false.	There is x for which $P(x)$ is true.
<u>$\neg \forall x P(x)$</u>	<u>$\exists x \neg P(x)$</u>	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

The reasoning in the table shows that:

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$
$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

These are important. You will use these.

Translation from English to Logic

Examples:

1. “**Some** student in this class has visited Mexico.”

Solution: Let $M(x)$ denote “ x has visited Mexico” and $S(x)$ denote “ x is a student in this class,” and U be all people.

$$\exists X (S(X) \wedge M(X))$$

2. “**Every** student in this class has visited Canada or Mexico.”

Solution: Add $C(x)$ denoting “ x has visited Canada.”

$$\forall x (S(x) \rightarrow (C(x) \vee M(x)))$$

Some Fun with Translating from English into Logical Expressions₁

$U = \{\text{fleegles, snurds, thingamabobs}\}$

$F(x)$: x is a fleegle

$S(x)$: x is a snurd

$T(x)$: x is a thingamabob

Translate “**Everything is a fleegle**”

Solution: $\forall x F(x)$

Some Fun with Translating from English into Logical Expressions.

$U = \{\text{fleegles, snurds, thingamabobs}\}$

$F(x)$: x is a fleegle

$S(x)$: x is a snurd

$T(x)$: x is a thingamabob

“Nothing is a snurd.”

Sol: $\forall x (\neg S(x)) \equiv \neg (\exists x S(x))$

Some Fun with Translating from English into Logical Expressions₃

$U = \{\text{fleegles, snurds, thingamabobs}\}$

$F(x)$: x is a fleegle

$S(x)$: x is a snurd

$T(x)$: x is a thingamabob

“All fleegles are snurds.”

Solution: $\forall X (F(X) \rightarrow S(X))$

Some Fun with Translating from English into Logical Expressions₄

$U = \{\text{fleegles, snurds, thingamabobs}\}$

$F(x)$: x is a fleegle

$S(x)$: x is a snurd

$T(x)$: x is a thingamabob

“Some fleegles are thingamabobs.”

注意, 用 $\rightarrow$ 是不对的

Solution: $\exists X (F(X) \wedge T(X))$

Some Fun with Translating from English into Logical Expressions₅

$U = \{\text{fleegles, snurds, thingamabobs}\}$

$F(x)$: x is a fleegle

$S(x)$: x is a snurd

$T(x)$: x is a thingamabob

“No snurd is a thingamabob.”

Solution: $\neg \exists X (S(X) \wedge T(X))$ What is this equivalent to?

$$\begin{aligned} & \equiv \\ & \forall X ((\neg S(X)) \vee (\neg T(X))) \equiv \forall X (S(X) \rightarrow \neg T(X)) \\ & \text{<key> } p \rightarrow q \equiv (\neg p) \vee q \end{aligned}$$

Some Fun with Translating from English into Logical Expressions₆

$U = \{\text{fleegles, snurds, thingamabobs}\}$

$F(x)$: x is a fleegle

$S(x)$: x is a snurd

$T(x)$: x is a thingamabob

“If any fleegle is a snurd then it is also a thingamabob.”

Solution: $\forall X ((F(X) \wedge S(X)) \rightarrow T(X))$ ✓

System Specification Example

Predicate logic is used for specifying properties that systems must satisfy.

For example, translate into predicate logic:

- “Every mail message larger than one megabyte will be compressed.”
- “If a user is active, at least one network link will be available.”

Decide on predicates and domains (left implicit here) for the variables:

- Let $L(m, y)$ be “Mail message m is larger than y megabytes.”
- Let $C(m)$ denote “Mail message m will be compressed.”
- Let $A(u)$ represent “User u is active.”
- Let $S(n, x)$ represent “Network link n is state x .”

Now we have: $\forall m (L(m, 1) \rightarrow C(m))$

$$\exists u A(u) \rightarrow \exists n S(n, \text{available})$$

Lewis Carroll Example



Charles Lutwidge
Dodgson (AKA
Lewis Carroll)
(1832-1898)

The first two are called *premises* (前提) and the third is called the *conclusion*.

1. “All lions are fierce.”
2. “Some lions do not drink coffee.”
3. “Some fierce creatures do not drink coffee.”

Here is one way to translate these statements to predicate logic.

Let $P(x)$, $Q(x)$, and $R(x)$ be the propositional functions “ x is a lion,” “ x is fierce,” and “ x drinks coffee,” respectively.

- | | | |
|----|-------------------------------------|---------------------------|
| 1. | $\forall X (P(X) \rightarrow Q(X))$ | $P(x)$: “ x is a lion” |
| 1. | $\exists X (P(X) \wedge \neg R(X))$ | $Q(x)$: “ x is fierce” |
| 1. | $\exists X (Q(X) \wedge \neg R(X))$ | $R(x)$: “ x drinks |

Later we will see how to prove that the conclusion follows from the premises

Chapter 1.

The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Nested Quantifiers

Nested (巢狀) quantifiers are often necessary to express the meaning of sentences in English as well as important concepts in computer science and mathematics.

Example: “**Every real number has an inverse**” is
 對於所有x, 有部分的y 讓 $x+y=0$ (y是x的 inverse)

$$\forall x (\exists y (x + y = 0))$$

where the domains of x and y are the real numbers.

We can also think of nested propositional functions:

$\forall x \exists y (x + y = 0)$ can be viewed as $\forall x Q(x)$ where $Q(x)$ is $\exists y P(x, y)$ where $P(x, y)$ is $(x + y = 0)$

Thinking of Nested Quantification

Nested Loops

- To see if $\forall x \exists y P(x,y)$ is true, loop through the values of x :
 - At each step, loop through the values for y .
 - **If for some pair of x and y , $P(x,y)$ is false, then $\forall x \exists y P(x,y)$ is false** and both the outer and inner loop terminate.

$\forall x \exists y P(x,y)$ is true if the outer loop ends after stepping through each x .

- To see if $\exists x \forall y P(x,y)$ is true, loop through the values of x :
 - At each step, loop through the values for y .
 - The inner loop ends when a pair x and y is found such that $P(x, y)$ is true.
 - **If no y is found** such that $P(x, y)$ is true the outer loop terminates as $\exists x \forall y P(x,y)$ **has been shown to be false.**

$\exists x \forall y P(x,y)$ is true if the outer loop ends after stepping through each x .

If the domains of the variables are infinite, then this process can not actually be carried out.

Order (順序) of Quantifiers

$$3. R(x, y) = x \cdot y = 0$$

Examples: $\exists y (\forall x R(x, y))$ value? = True
 $y=0$ 時, 對於所有 x , $x \cdot y = 0$

1. Let $P(x, y)$ be the statement “ $x + y = y + x$.” Assume that U is the **real numbers**. Then $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ have the same truth value. True

2. Let $Q(x, y)$ be the statement “ $x + y = 0$.” Assume that U is the real numbers. Then $\forall x (\exists y Q(x, y))$ is true, but

$\exists y (\forall x Q(x, y))$ is false.

對於一個 y 來說
 所有的 x 都滿足 $x + y = 0$

對於所有 x 來說
 可以 find 一個 y 來使 $x + y = 0$

Questions on Order of Quantifiers₁

Example 1: Let U be the real numbers,

Define $P(x,y) : x \cdot y = 0$

What is the truth value of the following:

1. $\forall x (\forall y P(x,y))$ <key> 所有 x , 所有 y

Answer: False

2. $\forall x (\exists y P(x,y))$ <key> 对于所有 x , 有个 y 使 $x \cdot y = 0$
($y=0$)

Answer: True

3. $\exists x (\forall y P(x,y))$ <key> 对于所有 y , 有个 x 使 $x \cdot y = 0$
($x=0$)

Answer: True

4. $\exists x (\exists y P(x,y))$ <key> 对一些 x 来说
有一些 y 使 $x \cdot y = 0 \Rightarrow$ 必 True

Answer: True

Questions on Order of Quantifiers₂

Example 2: Let U be the real numbers,

Define $P(x, y) : x / y = 1$

What is the truth value of the following:

1. $\forall x \forall y P(x, y)$

Answer: False

2. $\forall x \exists y P(x, y)$

Answer: False

3. $\exists x \forall y P(x, y)$

Answer: False

4. $\exists x \exists y P(x, y)$

Answer: True

5. $R(x, y) : \frac{x}{y} = 0$

$\exists x \forall y R(x, y) \Rightarrow \text{FALSE}$

$x=0$ $y=1 \sim \infty$ $\checkmark$
but $x=0$ 時, $\frac{0}{0} = \text{未定義}$

<key> 對於所有 x 來說

都有一個 y 使 $\frac{x}{y} = 1$?

$\rightarrow \text{FALSE!}$ $x=0$ 的情況不合!

<key>

對於所有 y 來說, 都有一個 x 使 $\frac{x}{y} = 1$? $\rightarrow \text{FALSE}$

Quantifications of Two Variables

Statement	When True?	When False
$\forall x \forall y P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall y \forall x P(x, y)$		
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$		
$\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y

Translating Nested Quantifiers into English

Example 1: Translate the statement

$$\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

where **C(x)** is “x has a computer,” and **F(x,y)** is “x and y are friends,” and the **domain** for both x and y consists of **all students in your school**.

Solution: Every student in your school (has a computer) or (has a friend who has a computer.)

$C(x) \wedge F(x, y)$ $C(x)$

Example 2: Translate the statement

$$\exists x \forall y \forall z ((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z))$$

Solution: There is a student none of whose friends are also friends with each other.

Translating Mathematical Statements into Predicate Logic

Example : Translate “The sum of two positive integers is always positive” into a logical expression.

Solution:

1. Rewrite the statement to make the implied quantifiers and domains explicit:
“For every two integers, if these integers are both positive, then the sum of these integers is positive.”
2. Introduce the variables x and y , and specify the domain, to obtain:
“For all positive integers x and y , $x + y$ is positive.”
3. The result is:

$$\forall x \forall y ((x > 0) \wedge (y > 0)) \rightarrow (x + y > 0)$$

where the **domain** of both variables consists of **all integers**

Translating English into Logical Expressions

Example: Use quantifiers to express the statement
 “There is a woman who has taken a flight on every airline in the world.”

Solution:

1. Let $P(w, f)$ be “ w has taken f ” and $Q(f, a)$ be “ f is a flight on a .”
2. The domain of w is all women, the domain of f is all flights, and the domain of a is all airlines.
3. Then the statement can be expressed as:

$$\exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$$

89 Questions on Translation from English

Choose the obvious predicates and express in predicate logic.

Example 1: “Brothers are siblings.”

Solution: $\forall x \forall y (B(x,y) \rightarrow S(x,y))$

Example 2: “Siblinghood is symmetric.”

Solution: $\forall x \forall y (S(x,y) \rightarrow S(y,x))$

Example 3: “Everybody loves somebody.”

Solution: $\forall x \exists y L(x,y)$

Example 4: “There is someone who is loved by everyone.”

Solution: $\exists y \forall x L(x,y)$

Example 5: “There is someone who loves someone.”

Solution: $\exists x \exists y L(x,y)$

Example 6: “Everyone loves himself”

Solution: $\forall x L(x,x)$

Negating Nested Quantifiers

Example 1: Recall the logical expression developed three slides back:

$$\exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$$

Part 1: Use quantifiers to express the statement that “There does **not** exist a woman who has taken a flight on every airline in the world.”

Solution: $\neg \exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$

Part 2: Now use De Morgan’s Laws to move the negation as far inwards as possible.



Solution:

$$\begin{array}{ll}
 1 & \neg \exists w \forall a \exists f (P(w, f) \wedge Q(f, a)) \\
 1 & \forall w \neg \forall a \exists f (P(w, f) \wedge Q(f, a)) \quad \text{by De Morgan's for } \exists \\
 1 & \forall w \exists a \neg \exists f (P(w, f) \wedge Q(f, a)) \quad \text{by De Morgan's for } \forall \\
 1 & \forall w \exists a \forall f \neg (P(w, f) \wedge Q(f, a)) \quad \text{by De Morgan's for } \exists \\
 1 & \forall w \exists a \forall f (\neg P(w, f) \vee \neg Q(f, a)) \quad \text{by De Morgan's for } \wedge.
 \end{array}$$

Part 3: Can you translate the result back into English?

Solution:



“For every woman there is an airline such that for all flights, this woman has not taken that flight or that flight is not on this airline”

Chapter 1.

The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 ~~Nested~~ Quantifiers

1.6 Rules of Inference (推理)

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Revisiting the Socrates (蘇格拉底) Example

We have the two premises:

- “All men are mortal.”
- “Socrates is a man.”

And the conclusion:

- “Socrates is mortal.”

How do we get the conclusion from the premises?

The Argument (論證)

We can express the **premises (above the line)** and the **conclusion (below the line)** in predicate logic as an argument:

$$\begin{array}{c}
 \text{人} \quad \text{會死} \\
 \forall x (Man(x) \rightarrow Mortal(x)) \\
 \hline
 \text{人} \quad Man(Socrates) \\
 \hline
 \therefore Mortal(Socrates) \\
 \text{會死} \quad \text{阿蘇}
 \end{array}$$

We will see shortly that this is a valid argument.

Valid Arguments₁

有效的

We will show how to construct valid arguments in two stages; first for **propositional logic** and then for **predicate logic**. The rules of inference are the essential building block in the construction of valid arguments.

1. **Propositional Logic**

Inference Rules

2. **Predicate Logic**

Inference rules for propositional logic plus additional **inference rules** to handle variables and quantifiers.

Arguments in Propositional Logic

A **argument** (論證) in propositional logic is a sequence of propositions. All but the final proposition are called **premises** (前提). The last statement is the **conclusion** (結論).

The argument is **valid** (有效) if the premises imply the conclusion. An **argument form** (論證形式) is an argument that is valid no matter what propositions are substituted into its propositional variables.

If the premises are $p_1, p_2, \dots, p_n$ and the conclusion is q then $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a **tautology** (恆真).

Rules of inference (推論規則) are all the simple argument forms that will be used to construct more complex argument forms.

Rules of Inference for Propositional

Logic: **Modus Ponens (前項肯定)**

☆

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Corresponding Tautology:

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

$$\frac{p \quad p \rightarrow q}{\therefore q}$$

Example.

注意
Truth Table

Let p be “It is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.” $p \rightarrow q$

“It is snowing.” p

→ “Therefore, I will study discrete math.”

Modus Tollens (後項否定)

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Corresponding Tautology:

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$$

$$\frac{\neg q \quad p \rightarrow q}{\therefore \neg p}$$

Example:

— Let p be “it is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.” $p \rightarrow q$

“I will not study discrete math.” $\neg q$

“Therefore, it is not snowing.”

Hypothetical Syllogism (假言三段論)

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Corresponding Tautology:

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Let p be “it snows.”

Let q be “I will study discrete math.”

Let r be “I will get an A.”

“If it snows, then I will study discrete math.”

“If I study discrete math, I will get an A.”

“Therefore, If it snows, I will get an A.”

Disjunctive Syllogism (選言三段論)

p	q	$\neg p$	$p \vee q$
T	T	F	T
T	F	F	T
F	T	T	T
F	F	T	F

Corresponding Tautology:

$$(\neg p \wedge (p \vee q)) \rightarrow q$$

$$p \vee q$$

$$\frac{\neg p}{\therefore q}$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math **or** I will study English literature.”

“I will **not** study discrete math.”

“Therefore, I will study English literature.”

Addition (加入)

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Corresponding Tautology:

$$p \rightarrow (p \vee q)$$

$$\frac{p}{\therefore p \vee q}$$

Example:

Let p be “I will study discrete math.”

Let q be “I will visit Las Vegas.”

“I will study discrete math.”

“Therefore, I will study discrete math **or** I will visit Las Vegas.”

Simplification (簡化)

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Corresponding Tautology:

$$(p \wedge q) \rightarrow p$$

$$\frac{p \wedge q}{\therefore p}$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math and English literature”

“Therefore, I will study discrete math.”

Conjunction (並言)

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Corresponding Tautology:

$$((p) \wedge (q)) \rightarrow (p \wedge q)$$

$$\frac{p}{q} \therefore p \wedge q$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math.”

“I will study English literature.”

“Therefore, I will study discrete math **and** I will study English literature.”

Resolution (歸結)

p	q	r	$p \vee q$	$q \vee r$	$\neg p \vee r$
T	T	T	T	T	T
T	T	F	T	T	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	F	T	T
F	F	F	F	F	T

Resolution plays an important role in AI and is used in Prolog.

Corresponding

Tautology:
 $((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$

$\neg p \vee r$

$p \vee q$

$\therefore q \wedge r$

Let p be “I will study discrete math.”

Let r be “I will study English literature.”

Let q be “I will study databases.”

“I will not study discrete math or I will study English literature.”

“I will study discrete math or I will study databases.”

“Therefore, I will study databases or I will study English

¹⁰⁴Using the Rules of Inference to Build Valid Arguments

A *valid argument* (有效論證) is a sequence of statements. Each statement is either a premise or follows from previous statements by rules of inference. The last statement is called **conclusion.**

A valid argument takes the following form:

S_1

S_2

.

.

.

S_n

$\square C$

Valid Arguments₂

Example 1: From the single proposition

$$p \wedge (p \rightarrow q)$$

Show that q is a conclusion.

Solution:

Step	Reason
1. $p \wedge (p \rightarrow q)$	Premise (前提)
2. p	Simplification using (1)
3. $p \rightarrow q$	Simplification using (1)
4. q	Modus Ponens using (2) and (3)

Valid Arguments₃

Example 2:

(假论)

With these hypotheses (premises):

“It is **not sunny this afternoon** and **it is colder than yesterday**.”

“**We will go swimming** only if **it is sunny this afternoon** .”

“If **we do not go swimming**, then **we will take a canoe trip**.”

“If **we take a canoe trip**, then **we will be home by sunset**.”

Using the inference rules, construct a valid argument for the conclusion:

“We will be home by sunset.”

Solution: ■

Choose propositional variables:

p : “It is sunny this afternoon.” r : “We will go swimming.” t : “We will be home by sunset.”

q : “It is colder than yesterday.” s : “We will take a canoe trip.”

Translation into propositional logic:

Hypotheses: $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$

Conclusion: t

Valid Arguments₄

Hypotheses: $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$

Conclusion: t

1. Construct the Valid Argument

Step

Reason

1. $\neg p \wedge q$

Premise

2. $\neg p$

Simplification using (1)

3. $r \rightarrow p$

Premise

4. $\neg r$

Modus tollens using (2) and (3)

5. $\neg r \rightarrow s$

Premise

6. s

Modus ponens using (4) and (5)

7. $s \rightarrow t$

Premise

8. t

Modus ponens using (6) and (7)

Handling Quantified Statements

Valid arguments (有效論證) for quantified statements are a sequence of statements. Each statement is either a premise or follows from previous statements by rules of inference which include:

- Rules of Inference for **Propositional Logic**
- Rules of Inference for **Quantified Statements**

The rules of inference (推論規則) for quantified statements are introduced in the next several slides.

Universal Instantiation (UI)

(全稱個例化規則)

$$\frac{\forall x P(x)}{\therefore P(c)}$$

因為都对，
所以假設 Domain 有 c
則 c 也會對！

Example:

Our domain consists of all dogs and Fido is a dog.

“All dogs are cuddly.”

“Therefore, Fido is cuddly.”

Universal Generalization (UG)

(全稱通則化規則)

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$$

前面(UI)的
反過來！

Used often implicitly in Mathematical Proofs.

Existential Instantiation (EI)

(存在個例化規則)

$$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$$

Example:

—
“There is someone who got an A in the course.”

“Let’s call her *a* and say that *a* got an A”

Existential Generalization (EG)

(存在通則化規則)

$$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$$

Example:

“Michelle got an A in the class.”

“Therefore, someone got an A in the class.”

Using Rules of Inference₁

Example 1: Using the rules of inference, construct a valid argument to show that

“John Smith has two legs”

is a consequence of the premises:

“Every man has two legs.” “John Smith is a man.”

Solution: Let $M(x)$ denote “ x is a man” and $L(x)$ “ x has two legs” and let John Smith be a member of the domain.

Valid Argument:

Step	Reason
1. $\forall x (M(x) \rightarrow L(x))$	Premise
2. $M(J) \rightarrow L(J)$	UI from (1)
3. $M(J)$	Premise
4. $L(J)$	Modus Ponens using (2) and (3)

Using Rules of Inference₂

Example 2: Use the rules of inference to construct a valid argument showing that the conclusion

“Someone who passed the first exam has not read the book.”

follows from the premises

“A student in this class has not read the book.”

“Everyone in this class passed the first exam.”

Solution: Let

$C(x)$ denote “ x is in this class”

$B(x)$ denote “ x has read the book”

$P(x)$ denote “ x passed the first exam”

$$\frac{\begin{array}{l} \exists x (C(x) \wedge \neg B(x)) \\ \forall x (C(x) \rightarrow P(x)) \end{array}}{\therefore \exists x (P(x) \wedge \neg B(x))}$$



如何得到此結論？

First we translate the premises and conclusion into symbolic form.

Continued on next slide □

Using Rules of Inference₃

如何得到此結論？

Valid Argument:

Step	Reason
1. $\exists x (C(x) \wedge \neg B(x))$	Premise
2. $C(a) \wedge \neg B(a)$	EI from (1)
3. $C(a)$	Simplification from (2)
4. $\forall x (C(x) \rightarrow P(x))$	Premise
5. $C(a) \rightarrow P(a)$	UI from (4)
6. $P(a)$	MP from (3) and (5)
7. $\neg B(a)$	Simplification from (2)
8. $P(a) \wedge \neg B(a)$	Conj from (6) and (7)
9. $\exists x (P(x) \wedge \neg B(x))$	EG from (8)

¹¹⁶ Returning to the Socrates Example

$$\frac{\forall x (Man(x) \rightarrow Mortal(x)) \quad Man(Socrates)}{\therefore Mortal(Socrates)}$$


如何得到此結論？

Solution for Socrates Example

如何得到此結論？

Valid Argument

Step	Reason
1. $\forall x (Man(x) \rightarrow Mortal(x))$	Premise
2. $Mam(Socrates) \rightarrow Mortal(Socrates)$	UI from (1)
3. $Mam(Socrates)$	Premise
4. $Mortal(Socrates)$	MP from (2) and (3)

Universal Modus Ponens

(全稱前項肯定)

Universal Modus Ponens combines universal instantiation and modus ponens into one rule.

$$\forall x (P(x) \rightarrow Q(x))$$

$P(a)$, where a is a particular
element in the domain

$$\therefore Q(a)$$

This rule could be used in the **Socrates example**.

Chapter 1.

The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs (前面所有章节的應用)

1.8 Proof Methods and Strategy

Proofs of Mathematical Statements

A *proof* (證明) is a valid argument that establishes the truth of a statement.

In math, CS, and other disciplines, informal proofs which are generally shorter, are generally used.

- More than one rule of inference are often used in a step.
- Steps may be skipped.
- The rules of inference used are not explicitly stated.
- Easier for to understand and to explain to people.
- But it is also easier to introduce errors.

Proofs have many practical applications:

- verification that computer programs are correct
- establishing that operating systems are secure
- enabling programs to make inferences in artificial intelligence
- showing that system specifications are consistent

Definitions

重点



A **theorem** (定理) is a statement that can be shown to be true using:

- definitions
- other theorems
- **axioms** (公理) (statements which are given as true)
- rules of inference

A **lemma** (引理) is a 'helping theorem' or a result which is needed to prove a theorem. lemma 去引領定理

A **corollary** (推論) is a result which follows directly from a theorem.
 '有了 Theorem, 引申而來 Corollary

Less important theorems are sometimes called **propositions** (命題).

A **conjecture** (推測) is a statement that is being proposed to be true.
Once a proof of a conjecture is found, it becomes a theorem. It may turn out to be false.

Forms of Theorems

聲稱

Many theorems assert that a property holds for all elements in a domain, such as the integers, the real numbers, or some of the discrete structures that we will study in this class.

Often the universal quantifier (needed for a precise statement of a theorem) is omitted by standard mathematical convention.

For example, the statement:

“If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$ ”

really means

“For all positive real numbers x and y , if $x > y$, then $x^2 > y^2$.”

Proving Theorems

Many theorems have the form:

$$\forall x (P(x) \rightarrow Q(x))$$

—

To show that where c is an arbitrary element of the domain,

$$P(c) \rightarrow Q(c)$$

By universal generalization the truth of the original formula follows.

So, we must prove something of the form: $p \rightarrow q$

¹²⁴ Proving Conditional Statements: $p \rightarrow q$

Trivial Proof (微不足道的證明): If we know q is true, then

瑣碎

$p \rightarrow q$ is true as well.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

“If it is raining then $1=1$.”

Vacuous Proof (空的證明): If we know p is false then

$p \rightarrow q$ is true as well.

“If I am both rich and poor then $2 + 2 = 5$.”

← 根據定義，
前項為 false，則結論必 True

[Even though these examples seem silly, both trivial and vacuous proofs are often used in mathematical induction, as we will see in Chapter 5)]

Even and Odd Integers

☆ **Definition:** The integer n is even if there exists an integer k such that $n = 2k$, and n is odd if there exists an integer k , such that $n = 2k + 1$. Note that every integer is either even or odd and no integer is both even and odd.

We will need this basic fact about the integers in some of the example proofs to follow. We will learn more about the integers in Chapter 4.

Proving Conditional Statements: $p \rightarrow q$

假設

Direct Proof (直接證明): Assume that p is true. Use rules of inference, axioms, and logical equivalences to show that q must also be true.

Example: Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”
↗ p ↘ q

Solution: Assume that n is odd. Then $n = 2k + 1$ for an integer k .
by definition
 Squaring both sides of the equation, we get:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2r + 1,$$

where $r = 2k^2 + 2k$, an integer.

We have proved that if n is an odd integer, then n^2 is an odd integer.
 (marks the end of the proof. Sometimes **QED** is used instead.)

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Proving Conditional Statements: $p \rightarrow q$

有理數

Definition: The real number r is **rational** if there exist integers p and q where $q \neq 0$ such that $r = p/q$

Example: Prove that the sum of two rational numbers is rational.

Solution: 假設 Assume r and s are two rational numbers. Then there must be integers p, q and also t, u such that

$$r = p/q, \quad s = t/u, \quad u \neq 0, \quad q \neq 0$$

$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu + qt}{qu} = \frac{v}{w} \quad \text{where } v = pu + qt$$

$w = qu \neq 0$

整數

$qu \neq 0$

Thus the sum is rational.

$\because$ by definition, $u \neq 0, q \neq 0$

Proving Conditional Statements: $p \rightarrow q$

→ 假設不是 p ($\neg p$) 則不是 q ($\neg q$)

Proof by Contraposition (逆否證明法): Assume $\neg q$ and show $\neg p$ is true also. This is sometimes called an **indirect proof (間接證明)** method. If we give a direct proof of $\neg q \rightarrow \neg p$ then we have a proof of $p \rightarrow q$.

Why does this work?

$\exists k$ 是整數
 $3n+2 = 2k+1$

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Example: Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution: Assume n is even. So, $n = 2k$ for some integer k . Thus

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1) = 2j \text{ for } j = 3k + 1$$

Therefore $3n + 2$ is even. Since we have shown $\neg q \rightarrow \neg p$, $p \rightarrow q$ must hold as well. If n is an integer and $3n + 2$ is odd (not even), then n is odd (not even).

129 Proving Conditional Statements: $p \rightarrow q$

4

Example: Prove that for an integer n , if n^2 is odd, then n is odd.

Solution: Use proof by contraposition. **Assume n is even** (i.e., not odd). Therefore, there exists an integer k such that $n = 2k$. Hence,

$$n^2 = 4k^2 = 2(2k^2)$$

and n^2 is even (i.e., not odd).

We have shown that if n is an even integer, then n^2 is even. Therefore by contraposition, for an integer n , if n^2 is odd, then n is odd.

Proving Conditional Statements: $p \rightarrow q$

Proof by Contradiction (反證法):

$$p \rightarrow q$$

"p is true, $\neg p$ is true"

製造与前提違背的事實!

To prove p , assume $\neg p$ and derive a contradiction such as $p \wedge \neg p$. (an indirect form of proof). Since we have shown that $\neg p \rightarrow F$ is true, it follows that the contrapositive $T \rightarrow p$ also holds.

Example: Prove that if you pick 22 days from the calendar, at least 4 must fall on the same day of the week (至少4天都是星期x).

Solution: Assume that no more than 3 of the 22 days fall on the same day of the week. Because there are 7 days of the week, we could only have picked 21 days. This contradicts the assumption that we have picked 22 days.

Theorems that are Biconditional Statements

To prove a theorem that is a biconditional statement, that is, a statement of the form $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true.

Example: Prove the theorem: “If n is an integer, then n is odd **if and only** if n^2 is odd.”

Solution: We have to show that both $p \rightarrow q$ and $q \rightarrow p$. Therefore we can conclude $p \leftrightarrow q$.

Sometimes **iff** is used as an abbreviation for “**if and only if**,” as in

“If n is an integer, then n is odd **iff** n^2 is odd.”

What is wrong with this?

“Proof” that $1 = 2$

Step

Reason

1. $a = b$

Premise

2. $a^2 = a \times b$

左右同乘 a

Multiply both sides of (1) by a

3. $a^2 - b^2 = a \times b - b^2$

同減 b^2

Subtract b^2 from both sides of (2)

4. $(a - b)(a + b) = b(a - b)$

Algebra on (3)

5. $a + b = b$

$a = b$

Divide both sides by $a - b$

6. $2b = b$

$a - b = 0$

但4→5時

Replace a by b in (5) because $a = b$

7. $2 = 1$

把 $(a-b)$

約分了

Divide both sides of (6) by b

Solution: Step 5. $a - b = 0$ by the premise and division by 0 is undefined.

Looking Ahead

If direct methods of proof do not work

- We may need a clever use of a proof by **contraposition**.
- Or a proof by **contradiction**.
- In the next section, we will see strategies that can be used when straightforward approaches do not work.
- In Chapter 5, we will see **mathematical induction (數學歸納法)** and related techniques.
- In Chapter 6, we will see combinatorial proofs

Chapter 1.

The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Proof by Cases₁

To prove a conditional statement of the form:

$$(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q$$

Use the tautology

$$\begin{aligned} &[(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q] \Leftrightarrow \\ &[(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)] \end{aligned}$$

Each of the implications $p_i \rightarrow q$ is a *case*.

Proof by Cases₂

Example: Let $a @ b = \max\{a, b\} = a$ if $a \geq b$, otherwise $a @ b = \max\{a, b\} = b$.

Show that for all real numbers a, b, c

$$(a @ b) @ c = a @ (b @ c)$$

(This means the operation $@$ is associative.)

Proof: Let a, b , and c be arbitrary real numbers.

Then one of the following 6 cases must hold.

1. $a \geq b \geq c$
2. $a \geq c \geq b$
3. $b \geq a \geq c$
4. $b \geq c \geq a$
5. $c \geq a \geq b$
6. $c \geq b \geq a$

Continued on next slide →

Proof by Cases₃

Case 1: $a \geq b \geq c$

$$(a @ b) = a, a @ c = \underline{a}, b @ c = b$$

$$\text{Hence } \underline{(a @ b) @ c} = \underline{a} = \underline{a @ (b @ c)}$$

Therefore the equality holds for the first case.

A complete proof requires that the equality be shown to hold for all 6 cases. But the proofs of the remaining cases are similar. Try them.

Without Loss of Generality

(不失一般性) (換一種假設)

Example: Show that if x and y are integers and both $x \cdot y$ and $x + y$ are even, then both x and y are even.

Proof: Use a proof by contraposition. Suppose x and y are not both even. Then, one or both are odd. Without loss of generality, assume that x is odd. Then $x = 2m + 1$ for some integer m . 不妨假设……

Case 1: y is even. Then $y = 2n$ for some integer n , so

$$x + y = (2m + 1) + 2n = 2(m + n) + 1 \text{ is odd.}$$

Case 2: y is odd. Then $y = 2n + 1$ for some integer n , so

$$x \cdot y = (2m + 1)(2n + 1) = 2(2m \cdot n + m + n) + 1 \text{ is odd.}$$

We only cover the case where x is odd because the case where y is odd is similar. The use phrase **without loss of generality (WLOG)** indicates this.

Existence Proofs

Proof of theorems of the form $\exists x P(x)$

Constructive existence proof:

- **Find an explicit value of c , for which $P(c)$ is true.**
- Then $\exists x P(x)$ is true by Existential Generalization (EG). 「存在」→ 找到!

Example: Show that **there is** a positive integer that can be written as the sum of cubes of positive integers in two different ways:

Proof: 1729 is such a number since

$$1729 = 10^3 + 9^3 = 12^3 + 1^3$$

Nonconstructive Existence Proofs

In a ***nonconstructive*** existence proof, we assume no c exists which makes $P(c)$ true and derive a contradiction.

Example: Show that **there exist** irrational numbers x and y such that x^y is rational.

Proof: We know that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$. If it is rational, we have two irrational numbers x and y with x^y rational, namely $x = \sqrt{2}$ and $y = \sqrt{2}$. But if $\sqrt{2}^{\sqrt{2}}$ is irrational, then we can let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ so that $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{(\sqrt{2} \times \sqrt{2})} = (\sqrt{2})^2 = 2$.

Counterexamples (反例)

Recall $\exists x \neg P(x) \equiv \neg \forall x P(x)$.

To establish that $\neg \forall x P(x)$ is true (or $\forall x P(x)$ is false) find a c such that $\neg P(c)$ is true or $P(c)$ is false.

In this case c is called a **counterexample** to the assertion $\forall x P(x)$.

Example: “**Every** positive integer is the sum of the squares of 3 integers.” **The integer 7 is a counterexample.** So the claim is false.

Uniqueness Proofs

Some theorems assert the existence of a unique element with a particular property, $\exists!x P(x)$. The two parts of a *uniqueness proof* are

- **Existence (存在性)**: We show that an element x with the property exists.
- **Uniqueness (唯一性)**: We show that if $y \neq x$, then y does not have the property.

Example: Show that if a and b are real numbers and $a \neq 0$, then **there is a unique** real number r such that $ar + b = 0$.

Solution.

- **Existence:** The real number $r = -b/a$ is a solution of $ar + b = 0$ because $a(-b/a) + b = -b + b = 0$.
- **Uniqueness:** Suppose that s is a real number such that $as + b = 0$. Then $ar + b = as + b$, where $r = -b/a$. Subtracting b from both sides and dividing by a shows that $r = s$.

Proof Strategies for proving $p \rightarrow q$

Choose a method.

1. First try a **direct method** of proof.
2. If this does not work, try an indirect method (e.g., try to prove the **contrapositive**).
逆否證明

For whichever method you are trying, choose a strategy.

1. First try *forward reasoning*. Start with the axioms and known theorems and construct a sequence of steps that end in the conclusion. Start with p and prove q , or start with $\neg q$ and prove $\neg p$.
2. If this doesn't work, try *backward reasoning*. When trying to prove q , find a statement p that we can prove with the property $p \rightarrow q$.

Backward Reasoning

Example: Suppose that two people play a game taking turns removing, 1, 2, or 3 stones at a time from a pile that begins with 15 stones. The person who removes the last stone wins the game. **Show that the first player can win the game no matter what the second player does.**

Proof: Let n be the last step of the game.

Step n : Player1 can win if the pile contains 1,2, or 3 stones.

Step $n-1$: Player2 will have to leave such a pile if the pile that he/she is faced with has 4 stones.

Step $n-2$: Player1 can leave 4 stones when there are 5,6, or 7 stones left at the beginning of his/her turn.

Step $n-3$: Player2 must leave such a pile, if there are 8 stones .

Step $n-4$: Player1 has to have a pile with 9,10, or 11 stones to ensure that there are 8 left.

Step $n-5$: Player2 needs to be faced with 12 stones to be forced to leave 9,10, or 11.

Step $n-6$: Player1 can leave 12 stones by removing 3 stones.

Now reasoning forward, the first player can ensure a win by removing 3 stones and leaving 12.

Universally Quantified Assertions₁

To prove theorems of the form $\forall x P(x)$, assume x is an arbitrary member of the domain and show that $P(x)$ must be true. Using UG it follows that $\forall x P(x)$

Example: An integer x is even if and only if x^2 is even.

Solution: The quantified assertion is

$$\forall x [x \text{ is even} \leftrightarrow x^2 \text{ is even}]$$

We assume x is arbitrary.

Recall that $p \rightarrow q$ equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$

So, we have two cases to consider. These are considered in turn.

Continued on next slide

Universally Quantified Assertions₂

Case 1. We show that **if x is even** then x^2 is even using a **direct proof** (the *only if* part or *necessity*).

If x is even then $x = 2k$ for some integer k .

Hence $x^2 = 4k^2 = 2(2k^2)$ which is even since it is an integer divisible by 2.

This completes the proof of case 1.

Case 2 on next slide $\square$

¹⁴⁷ Universally Quantified Assertions₃

主張

Case 2. We show that **if x^2 is even** then x must be even (the *if* part or *sufficiency*). **We use a proof by contraposition.**

Assume x is not even and then show that x^2 is not even.

If x is not even then it must be odd. So, $x = 2k + 1$ for some k .

Then $x^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$

which is odd and hence not even. This completes the proof of case 2.

Since x was arbitrary, the result follows by UG.

Therefore we have shown that x is even if and only if x^2 is even.